Sathwik Munigala

B.Tech — Mechanical Engineering

IIITDM KURNOOL, Batch: 2023–2027

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Hyderabad, Telangana

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CAREER OBJECTIVE

Final-year B.Tech student in Mechanical Engineering with a minor in Robotics, with hands-on experience across the complete robot development cycle: mechanical design, electronics, simulation, reinforcement learning and deployment on real hardware. Seeking a core engineering role in robotics, mechanical design or automation.

EDUCATIONAL QUALIFICATIONS

Degree / Qualification	Institution	Year	Score / CGPA
B.Tech in Mechanical Engineering and Minor Robotics	Indian Institute of Information Technology Design & Manufacturing, Kurnool	2023–2027	7 / 10
Intermediate (MPC)	Narayana Junior College, Madhapur, Hyderabad	2021–2023	95.5%
Class X	Narayana High School, Tarnaka, Hyderabad	2021	10 / 10

TECHNICAL SKILLS & KNOWLEDGE

Robotics and Simulation	NVIDIA Isaac Sim, Isaac Lab, ROS 2 Humble, micro-ROS, Nav2, RViz, URDF and USD robot modelling
Robot Learning and Control	Reinforcement learning (PPO), reward design, curriculum learning, forward and inverse kinematics, Denavit-Hartenberg parameters, gait and foot trajectory design, sim-to-real transfer
Programming	Python, PyTorch, Embedded C, Linux (Ubuntu), Git
Embedded and Electronics	Raspberry Pi, ESP32, Arduino, PCA9685 PWM servo driver, I2C, servo calibration, power distribution and wiring
Design Software	SolidWorks, Fusion 360, Siemens NX, AutoCAD, MATLAB
Manufacturing	CNC Programming, Wire EDM, Welding Processes, Milling Machine, Turning, Metrology & QC, 3D Printing
Mechanical Design and Analysis	Machine design, mechanism design, kinematic analysis, gear design, tolerance analysis, GD&T, Design for Manufacturing (DFM), IS codes and BIS Standards
Computer Skills	MS Office (Excel, Word), Python, Engineering Report Writing
Domain Subjects	Kinematics of Machinery, Dynamics of Machinery, Machine Design, Heat Transfer

ACADEMIC PROJECTS

Quadruped Robot: Design, Learning-Based Walking and Sim-to-Real — Final Year Project 2026

Developed a 12-degree-of-freedom four-legged robot through the complete engineering cycle: mechanical design, robot modelling, electronics, simulation, control and transfer to the real hardware. Walking was developed in two ways. The first is learning-based, where a locomotion policy is trained with reinforcement learning (PPO) in NVIDIA Isaac Sim and Isaac Lab. The second is model-based, where a trot gait is generated from the kinematics of the legs and a learned policy keeps the robot balanced. The robot responds to forward, backward, sideways and turning commands, and the motion produced in simulation is reproduced on the real robot in real time.

In future, the robot will be extended to walk on rough terrain and climb stairs. Demonstration videos of the simulation-to-real transfer, showing the real robot and the simulation side by side, and further technical details are available on request.

Design and Fabrication of Manual Gear Transmission System — 3rd Year 2025–2026

Executed the complete end-to-end development of a manual transmission system. Performed kinematic analysis and

3D modelling in SolidWorks, followed by fabrication and assembly of the gears. Focused on gear ratio optimisation, synchronizer mechanism design, and testing structural integrity.

INTERNSHIP / INDUSTRIAL TRAINING

CAD Design Intern — Primeform Labs LLP May 2026 – Jun 2026, Hyderabad

- Executed the end-to-end product design cycle for industrial pressure valves and custom connectors, transitioning from initial rough conceptual sketches to final production-ready assemblies.
- Conducted analytical and mechanical calculations to determine optimal dimensions, wall thicknesses, and material safety factors meeting industry manufacturing standards.
- Developed high-fidelity 3D CAD models, robust parametric designs, and production-ready 2D CAD drawing files using strict DFM (Design for Manufacturing) guidelines to ensure flawless machine shop execution.

PERSONAL PROJECTS

Quadruped Robot Navigation (Research Project)

- Developed a simulated quadruped robot for autonomous navigation using the ROS2 Nav2 framework. Implemented localization and multi-planner navigation pipelines and conducted benchmarking analysis as part of an ongoing research article.
- **Extension:** migrated the pipeline to the latest NVIDIA Isaac Sim release and built a fully automated benchmarking loop that launches Isaac Sim, runs a planner end-to-end, then tears it down and relaunches with the next planner — sequencing all planner runs unattended with no manual intervention.

Motor Control via ROS2 and Micro-ROS (ESP32)

- Real-time DC motor control using Micro-ROS on embedded systems (ESP32/Arduino) with ROS 2 integration.

ANYmal-C Quadruped Locomotion Training (PPO + Isaac Gym)

- Trained the ANYmal-C quadruped robot to walk using Proximal Policy Optimisation (PPO) in the NVIDIA Isaac Gym physics simulator. Implemented actor-critic neural networks for robust locomotion control across various terrains.

ACHIEVEMENTS & CERTIFICATIONS

- Under publication — Technical Paper: "Comparative Analysis of Heterogeneous Planner Stack Structure for Robust Quadruped Navigation"

CO-CURRICULAR & EXTRA-CURRICULAR

- Robotics Club Head (Coordinator), 2024–2026
- Core Member — Organised technical workshops and mechanical fabrication sessions; mentored students in mechanical design and robotics system development.
- NSS Volunteer — Participated in blood donation, literacy drives, and disaster relief.